Poisoned data

AI security · Level 3 · Lesson 1



Today we will

- Explain how poisoned data and backdoors spoil a model.
- Find examples far from their group's median.
- Plan checks before new data goes into training.

Remember

- Wrong examples in, wrong model out: the attacker needs only the data.
- The median barely moves for one bad value: flag what sits far from it.
- Look before you delete, record what you removed, and keep a clean test set.

Find the poison

1. Warm-up (5 min): How would you teach a machine mango from lime?
2. Measure (10 min): Measure and label items of two kinds.
3. Poison (5 min): Swap three labels in secret.
4. Hunt (15 min): Find each median and flag far values.
5. Wrap-up (5 min): Caught, missed, honest flags; then the exit ticket.

No scale? Use lengths of leaves or sticks measured with a ruler, or print a table of 15 measurements. Choose two kinds whose values don't overlap much, so the swapped labels stand out.

Let's discuss

1. Why could the poison change the model even though nobody touched it?
2. Which honest items were flagged, and why?
3. What rules would keep our class data clean?

Exit ticket

- Write one way poison gets into data, and one check that would catch it.